Universal Plastic Shredder V2.0

Electrical Control & Power System

Portland State University

Maseeh College of Engineering & Computer Science

Department of Electrical & Computer Engineering

Final Report — Rough Draft

Team 07 — I'm So Shredded, Inc.

Bao Nguyen Fearghus Tyler Yaqoub Rabiah Fox Kang

Industry Sponsor: MME / Ichor Systems — Bridget Lannigan

Faculty Advisors: Prof. Andrew Greenberg • Dr. Jonathan Bird (EE) • Robert Paxton (ME)

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[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]

Contents

1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost $10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand deadman control, hardwired E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: $3,000 (EE portion approximately $1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., "shredder validated to X HP load with Y throughput rate") once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

2 Introduction

2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.

- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

3 Requirements

3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.
- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within $3,000 total project budget.

3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

4 System Architecture

4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.

- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** bench/test unit is a GE 5KE182BC205B (3 HP, 230 V low-voltage connection, 1750 RPM, FLA 7.6 A); its nameplate drives the VFD motor parameters. **[TODO: Final production motor selection is TBD (3–5 HP per requirements).]**
- **Brake resistor:** 75 Ω, 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop and two-hand deadman start; motor contactor energized through the safety chain. Access guarding at the feed is handled by the ME team's mechanical design, so no separate electrical lid interlock is used.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset) → PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU –V) → VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high) → PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI ↔ PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).

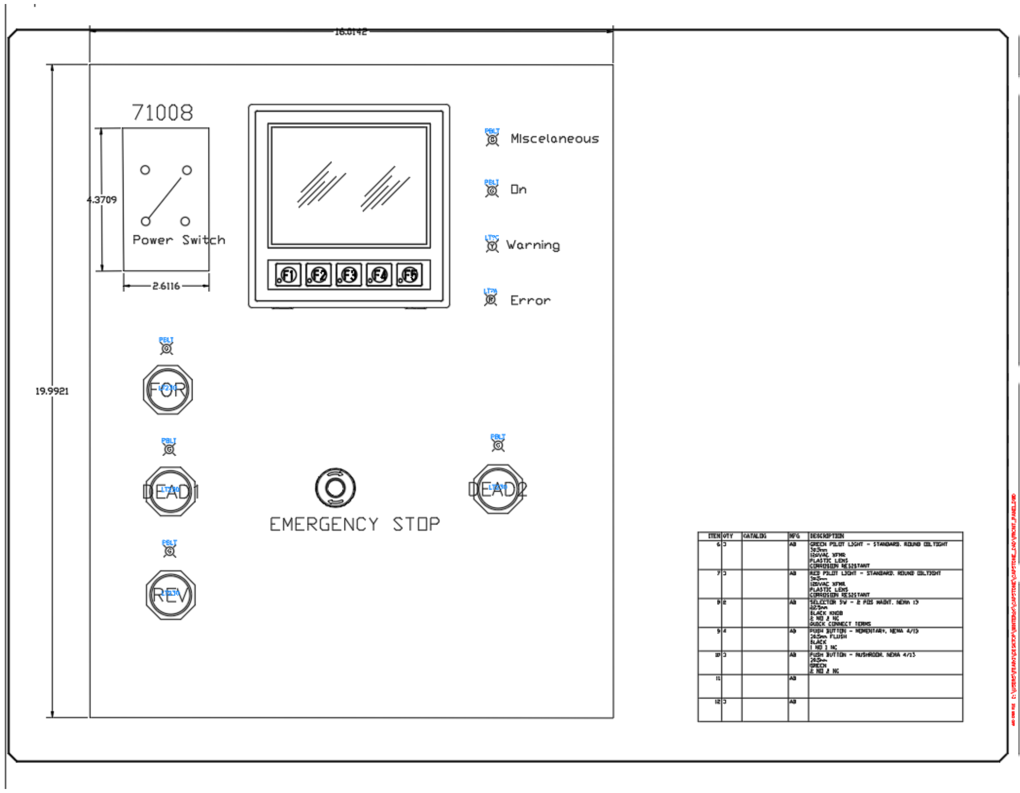

Figure 1: Operator control panel layout: HMI, power switch, forward/reverse and deadman pushbuttons, emergency stop, and status indicators.

[TODO: Add a system block diagram (power + signal flow) to complement the panel layout above.]

5 Hardware Design

5.1 Power Distribution

- Main disconnect / power switch: PowerTec 71008 two-pole (DPST) START/STOP switch, 120–230 VAC single-phase, rated 20 A / 3 HP max; switches both the line and neutral conductors.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90 °C ampacity columns of NEC Table 310.16 – well above the motor's 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG,

landed directly as straight stripped conductors at the terminal blocks (no ferrules used in this build).

- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

5.2 Motor and VFD

The Huanyang HY02D211B-T was selected over alternatives via a decision matrix (price, documentation, reliability). A brake resistor on the P/Pr terminals absorbs the regenerative energy from rapid reversal during the jam-clear cycle. The drive is programmed as shown in Table 1.

Table 1: Programmed VFD parameters (Huanyang HY02D211B-T).

Param	What it does	Value	Why this value
`PD141`	Motor rated voltage	230 V	Matches the GE test-motor nameplate (230 V low-voltage connection); sets the base of the V/Hz curve.
`PD142`	Motor rated current (FLA)	7.6 A	Nameplate full-load amps; sets the drive's current limit and electronic-overload reference.
`PD143`	Number of motor poles	4	Nameplate; 4-pole = 1800 RPM synchronous, so the drive resolves speed and slip correctly.
`PD144`	Motor rated speed	1750 RPM	Nameplate full-load speed; with the pole count it lets the drive estimate slip.
`PD002`	Run / frequency source	1 (external)	Speed comes from the external 0–10 V analog input (VI) driven by the PLC, not the keypad.
`PD052`	Relay output (FA/FC) function	02 (fault)	Closes the FA–FC relay on any fault so the PLC reads a jam/trip on input X5.

continued on next page . . .

Param	What it does	Value	Why this value
`PD054`	Analog output (VO) function	1 (current)	Outputs 0–10 V proportional to motor current to the PLC analog input for monitoring.
`PD123`	Over-torque action	3 (trip)	Over-torque self-trips the drive (which fires the fault relay); see the note below.
`PD124`	Over-torque threshold	150%	Set between no-load current (∼0.5 A) and measured jam current (>11 A) so normal shredding does not trip but a jam does.
`PD125`	Over-torque detection time	3.0 s	The over-torque must persist 3 s before tripping, so brief load spikes (feeding a piece) do not nuisance-trip.

[NOTE: The over-torque relay function (`PD052 = 12`) did not fire the relay in our specific Huanyang revision when `PD123 = 2` (continue running). Documented as a firmware-specific limitation (see Table 1). Working configuration uses `PD052 = 02` (fault indication) and `PD123 = 3` (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD's latched fault state between retry attempts.]

5.3 Safety Circuit

- E-stop: NC contact (E22B1) hardwired in series with the SD-N35 contactor coil. Pressing it drops the coil and opens the contactor, removing motor power independently of the PLC.
- Two-hand deadman: both buttons must be held continuously to enable the motor (evaluated by the PLC; see the Safety Functions assessment below).
- Access guarding: handled by the ME team's mechanical design (feed geometry), so no separate electrical lid interlock is used (this was a may-have, not a requirement).
- Motor contactor: the SD-N35 is energized through the hardwired safety chain, and the VFD is powered through it, so the safety chain has hardware precedence over the PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (`PD052 = 02`). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA–FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads `[ X5 ]` on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the independent hard-wired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

6 Software Design

6.1 PLC Ladder Logic Architecture

- Do-more Designer stage program (similar to grafcet / SFC).
- States: IDLE, RUNNING, JAM_DETECTED, RST_PULSE, REVERSE, RESUME, LOCKOUT.
- Transitions driven by deadman input, X5 jam signal, jam counter, and reset button.

6.2 State Machine

- **IDLE** → RUNNING when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → JAM_DETECTED on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM_DETECTED** → RST_PULSE if jam count < 3, → LOCKOUT if jam count ≥ 3.
- **RST_PULSE** (500 ms) → REVERSE (2 s).

- **REVERSE** → RESUME (brief delay) → back to RUNNING.
- **LOCKOUT** → IDLE only via dedicated reset button.

6.3 Jam Counter Reset Logic

- Counter increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets to 0 if motor runs cleanly for 30 continuous seconds in the RUNNING state.
- Also resets if deadman is released for 30 continuous seconds (session reset).
- Whichever condition fires first wins.

6.4 Startup Mask

- For first 5 seconds after PLC power-up, X5 is ignored.
- In NO active-high wiring, this guards against transient X5 pulses during VFD initialization. Less critical than under NC wiring but kept as a defensive measure.

6.5 HMI Programming

- Status displays: Ready, Running, Jam in progress, Lockout, Fault.
- Operator inputs: Speed setpoint (currently fixed in PLC), Reset button.
- Data displayed: Motor current (from the VFD VO analog output into PLC input F2-08AD-1 CH1 – wired but not yet correctly scaled, see Open Issues), Jam count this session, Total runtime.

[QUESTION: We have not finalized the HMI screen layout. Should we include mock-ups in the rough draft, or wait for the final?]

7 Safety Design

7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.

- Control wires land as straight stripped conductors at the terminal blocks (no ferrules used in this build).

7.2 Safety Functions

- **E-stop:** Category 0 stop (immediate removal of power). Hardwired — an E22B1 NC contact in series with the SD-N35 contactor coil; pressing it drops the coil and opens the contactor independently of the PLC. Single channel; the PLC additionally monitors the state on X4 (not a safety-rated channel).
- **Two-hand / deadman control:** both blue pushbuttons must be held continuously to enable the motor. As-built, the two NO inputs are AND-ed in the standard PLC's ladder logic to gate the VFD run output. It is the run-permissive and the safety interlock satisfying requirement M6, but it is *not* a safety-rated two-hand device per ISO 13851 Type III (no simultaneity timer, complementary contacts, or self-monitoring).
- **Access guarding:** handled by the ME team's mechanical design (feed geometry), so no electrical lid interlock is implemented (a lid interlock was a may-have, not a requirement).
- **Capacitive-touch module (planned):** a capacitive-touch safety module is on hand and the ladder logic to act on it (input X7 → stop motor + fault indication) is written, but it is *not yet wired*. This is a may-have (additional safety sensing) and is not relied upon in the current build.
- **Overload:** the VFD's electronic motor-overload function plus the over-torque trip (see Jam Detection); equipment protection rather than a personnel-safety function.

7.3 Performance Level (ISO 13849-1) — Preliminary Assessment

This is a preliminary, self-assessed determination from the as-built architecture. A formal verification needs a documented ISO 12100 risk assessment (to set the required performance level, PLr) and component reliability data (MTTFd / B10d, diagnostic coverage). Table 2 summarizes the achieved categories.

Table 2: Preliminary ISO 13849-1 assessment of the as-built safety functions.

Safety function	**As-built architecture**	**Category (prelim.)**
Emergency stop	Hardwired E22B1 NC contact in series with the SD-N35 contactor coil; opening it removes motor power. Single channel, well-tried components; PLC monitors X4 (not safety-rated).	Cat 1 (PL b–c)
Two-hand / deadman	Two NO buttons AND-ed in the standard PLC to gate the VFD run output. Single channel; no simultaneity check, complementary contacts, or self-monitoring.	Cat B (PL b); not ISO 13851 Type III
Over-torque / jam stop	VFD over-torque trip plus electronic motor overload. Equipment protection, not a personnel-safety function.	Not rated (process protection)

Overall the control system is approximately **Category B–1 (PL b–c)**, limited by: single-channel architecture throughout; a standard (non-safety) PLC evaluating the two-hand permissive; a single contactor with no force-guided (mechanically linked) contacts and no feedback (EDM) monitoring to detect a welded contactor; and the as-built neutral-switching deviation (see the NEC note). The hardwired E-stop, built from well-tried components, is the strongest function (Category 1).

To raise the achieved PL — for example to Category 3 / PL d, should the risk assessment require it for the rotating-blade hazard — would need a dual-channel architecture, a safety relay or safety-rated PLC, two contactors (or force-guided contacts) with feedback monitoring, switching of the hot leg, and a synchronous two-hand module meeting ISO 13851.

[TODO: Confirm with the advisor whether formal PL/SIL verification is in capstone scope. If so, document the target PLr from the ISO 12100 risk assessment and the component MTTFd / B10d and DC values to complete the calculation.]

7.4 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.

- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

8 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

8.1 Standards Summary Matrix

Standard	Scope	How it is applied here
NFPA 70 (NEC)	National Electrical Code – premises wiring, motor circuits	Branch-circuit, conductor, grounding, and disconnect design (see §8.2).
NFPA 79	Electrical standard for industrial machinery	Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §8.3).
NEMA 250	Enclosure type ratings	Control enclosure selected to a NEMA type matched to the lab/shop environment (see §8.4).
NEMA ICS 1/2	Industrial control & systems	Contactor / control device selection and marking.
NEMA MG 1	Motors and generators	Motor nameplate interpretation and VFD parameterization.

continued on next page . . .

Standard	Scope	How it is applied here
UL 508A	Industrial control panels	Panel construction practice and short-circuit current rating (SCCR) awareness.
IEC 60204-1	Safety of machinery – electrical equipment	International counterpart to NFPA 79; used to cross-check stop categories.
ISO 12100	Risk assessment / risk reduction	Framework behind the hazard analysis driving the safety circuit.
ISO 13849-1	Safety-related parts of control systems (PL)	Preliminary self-assessment (Cat B–1) of the E-stop and two-hand control chain.
ISO 13850	Emergency stop function	E-stop actuator type, color, and Category 0 behavior.
ISO 13851	Two-hand control devices	Deadman two-hand start timing and configuration.
OSHA 29 CFR 1910.147	Lockout/Tagout (LOTO)	Lockable main disconnect for energy isolation during service.

8.2 NFPA 70 – National Electrical Code

NEC article	Requirement	How it is applied here
Art. 430.6	Motor FLC basis	From the GE test-motor nameplate (FLA 7.6 A at 230 V); basis for all downstream sizing. Re-verified when the production motor is selected.
Art. 430.22	Branch conductor sizing	Conductors sized at $\geq$ 125% of motor FLC.
Art. 430.32	Running overload	VFD electronic motor-overload (`PD142 = 7.6 A`) plus the over-torque trip.
Art. 430.52	Branch short-circuit / ground-fault	30 A C-curve breaker on the VFD branch; clears inrush without nuisance trips (inverse-time-breaker allowance).
Art. 430.102/.109	Disconnecting means	Provided ahead of the controller and VFD.
Art. 430 Part X (430.122)	Adjustable-speed drive systems	Conductor and protection rules applied to the VFD-fed motor circuit.
Art. 250	Grounding & bonding	EGC (green / green-yellow) bonds the VFD, PSU, motor frame, and panel to a common ground bus and building earth.
Art. 310	Conductor ampacity	Conductors chosen from the ampacity tables for the installed temperature rating.
Art. 409	Industrial control panels	Panel layout, marking, and overcurrent considerations.
Art. 110.26	Working space	Clearances around live parts maintained for safe service. **[TODO: Confirm final panel mounting leaves required clearance.]**

[NOTE: As-built deviation flagged during the Point List review: the SD-N35 safety contactor switches the VFD neutral *(T) leg rather than the* hot *(R) leg. NEC / NFPA 79 practice is to switch the ungrounded (hot) conductor so that opening the contactor de-energizes the load to ground potential. Switching the grounded conductor leaves the load at line potential when the contactor is open. This is documented in the as-built I/O point list (Appendix B,*

VFD terminals) and must be corrected, or explicitly justified and risk-assessed, before final sign-off.]

8.3 NFPA 79 – Electrical Standard for Industrial Machinery

NFPA 79 clause	**Requirement**	**How it is applied here**
§5.3	Supply disconnecting means	Lockable main disconnect (PowerTec 71008) isolates the machine from supply; also satisfies LOTO.
§7	Overcurrent protection	Branch and downstream fusing at terminal blocks (PSU 25 A slow-blow, PLC 1 A fast-blow, control loads individually fused).
§8	Protective bonding circuit	Continuous equipment-grounding / bonding path to all exposed conductive parts.
§9	Control circuits & functions	Stop functions classified by stop category; the hard-wired safety chain has precedence over the PLC.
§10.7–10.8	Operator interface / E-stop	Red mushroom actuator on a yellow background, self-latching NC contact; start devices guarded by the two-hand requirement.
§13.2	Conductor color coding	Black AC line/load, red AC control, blue DC control, blue/white DC return, green/yellow PE; wires labeled both ends, landed straight (no ferrules in this build).

8.4 NEMA Standards

Standard	Covers	How it is applied here
NEMA 250	Enclosure type ratings	Targeting Type 12 (dust-tight, drip-tight) for the PLC/HMI panel. **[TODO: Confirm final enclosure type and dust-ingress handling.]**
NEMA ICS 1 / ICS 2	Industrial control & systems	Contactor and control-device selection/marking; the Mitsubishi SD-N35 is sized above motor FLA for AC-3 (motor) duty.
NEMA MG 1	Motors & generators	Interpreting the motor nameplate (voltage, poles, RPM, service factor) for VFD parameters `PD141--PD144`.

8.5 Machine-Safety Standards (ISO / IEC)

Standard	Covers	How it is applied here
ISO 12100	Risk assessment / reduction	Identifies the primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.
ISO 13849-1	Safety-related control parts (PL)	Preliminary assessment: hardwired E-stop $\approx$ Cat 1; two-hand permissive $\approx$ Cat B (see the Safety Functions section).
ISO 13850	Emergency stop function	Category 0 stop (immediate removal of power) with fail-safe NC wiring.
ISO 13851	Two-hand control devices	Both buttons actuated within 0.5 s of each other and held continuously to enable the motor.
IEC 60204-1	Electrical equipment of machines	International analogue of NFPA 79; used to cross-check stop categories and supply disconnection.

8.6 Occupational Safety (OSHA)

Regulation	Covers	How it is applied here
29 CFR 1910.147	Lockout/Tagout (LOTO)	The main supply disconnect is lockable for positive isolation during blade-area service and maintenance.
29 CFR 1910 Subpart S	Electrical safety	General electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build.

[TODO: Before final submission: convert any remaining "targets / intends to" language above into verified compliance statements, attach the panel enclosure datasheet, and resolve the NFPA 79 neutral-switching deviation note.]

9 Testing and Validation

9.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (`PD124 = 1`, `PD125 = 5.0`): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (`PD124 = 150`) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).

9.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

9.3 Safety Validation

- E-stop response time: **[TODO: measure with scope]**.
- Two-hand control timing: **[TODO: verify both buttons must be held continuously; simultaneous release stops within X ms]**.

10 Results

10.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with the PLC over the serial COM link.

10.2 Open Issues

- Vendor-specific Huanyang firmware quirk: `PD052 = 12` does not fire relay in continue-running mode. Working around with `PD052 = 02` + `PD123 = 3`.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.
- Motor-current monitor (VFD `VO` → F2-08AD-1 CH1) is wired, but under time constraints it was never configured/scaled correctly to read the VFD output current. The signal path exists; the VFD analog-output and PLC analog-input scaling still need to be set and calibrated.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

11 Discussion

11.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD ($150) over ATO ($450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.

- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NO active-high versus NC fail-safe jam signal: the VFD fault relay is wired NO active-high (FA–FC closes on trip, X5 HIGH). Simpler to wire, but not fail-safe at the signal level—a broken wire reads as "no fault." That residual risk is covered by the independent hardwired E-stop loop and motor contactor, which remove power regardless of the PLC.

11.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on control wires in future builds: this build landed straight stranded wire at the terminal blocks, and loose/spread strands caused multiple debugging hours during commissioning.

11.3 Project Management Notes

[TODO: Add a paragraph on how the team coordinated, what tools we used (GitHub, weekly meetings, etc.), and how schedule slipped or held.]

12 Future Work and Improvements

12.1 Safety and Compliance

- **Integrate the capacitive-touch safety (CTSI) module:** wire the on-hand capacitive-touch module to input X7. The ladder logic to stop the motor and raise a fault on a touch trip is already written, so this is primarily a wiring and validation task. It adds a no-contact operator-presence safeguard at the feed.
- **Raise the safety performance level:** move from the current single-channel Cat B–1 toward Cat 3 / PL d — a safety relay or safety-rated PLC, dual-channel inputs,

two contactors (or force-guided contacts) with feedback (EDM) monitoring, and a synchronous two-hand module per ISO 13851.

- **Correct the supply switching:** switch the hot (ungrounded) conductor through the contactor rather than the neutral, per NFPA 79 / NEC.
- **Ferrule the control wiring:** crimp ferrules on stranded control conductors at the terminal blocks to eliminate the loose-strand faults seen during commissioning.

12.2 Monitoring and Connectivity

- **Telemetry:** stream live operating data (motor current, speed, jam count, fault state, runtime) off the PLC for dashboards and trend analysis.
- **Internet / remote viewing:** expose a read-only view of device status over the network — via the H2-DM1E's built-in web server or an ESP32 / edge gateway — so the shredder can be monitored remotely from a browser or phone.
- **Data logging:** record total runtime, jam events per shift, and faults locally for predictive maintenance.

12.3 Capability

- **Predictive jam detection:** train a small model on the motor-current waveform shape just before a jam to detect and pre-empt jams earlier.
- **Mechanical integration:** add a feed hopper that auto-meters plastic into the blades.
- **Higher-capacity model:** scale to a 5–10 HP drivetrain for greater throughput.

13 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – layered safety functions (hardwired E-stop, two-hand deadman control), NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.
- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.

- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

14 References

- NFPA 70 (2023): National Electrical Code (esp. Articles 110, 250, 310, 409, 430).
- NFPA 79 (2024): Electrical Standard for Industrial Machinery.
- NEMA 250: Enclosures for Electrical Equipment.
- NEMA ICS 1 / ICS 2: Industrial Control and Systems.
- NEMA MG 1: Motors and Generators.
- UL 508A: Industrial Control Panels.
- IEC 60204-1:2016: Safety of Machinery – Electrical Equipment of Machines.
- ISO 12100:2010: Safety of Machinery – Risk Assessment and Risk Reduction.
- ISO 13849-1:2023: Safety-related parts of control systems.
- ISO 13850:2015: Emergency Stop Function.
- ISO 13851:2019: Two-hand control devices.
- OSHA 29 CFR 1910.147: The Control of Hazardous Energy (Lockout/Tagout).
- Huanyang HY-series VFD User Manual.
- AutomationDirect DL205 / Do-more User Manual.
- GE 5KE Frame 182T Motor Datasheet.

[TODO: Add full reference list with URLs / publication info before final.]

15 Appendices

15.1 Appendix A: Bill of Materials

Major purchased items for the electrical system (procurement BOM, rev 02/20/26):

Item	Description	Mfr. P/N	Critical
Motor	AC induction motor, 3-phase (bench-tested with GE 5KE182BC205B)	TBD	Yes
VFD	Huanyang 1-phase in / 3-phase out drive, 3 HP	HY02D211B-T	Yes
HMI	C-More Micro-Graphic 4" color touch panel	EA1-T4CL	Yes
Circuit breaker	DIN-rail branch breaker	NBE7	No
Power switch	Single-phase ON/OFF disconnect, 120–230 V	71008	No
Status lights	Red / green panel indicators	Ichor-provided	No
E-stop	Mushroom emergency-stop pushbutton	Ichor-provided	Yes

[NOTE: The final production motor is TBD (3–5 HP per requirements). Bench testing and the VFD motor parameters (PD141–144) use the GE 5KE182BC205B nameplate the team has on hand (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM); the production motor's nameplate will supersede those values once selected. Additional control-panel hardware (PLC rack/modules, Mean Well NDR-480-24 PSU, SD-N35 contactor, Phoenix Contact breaker, brake resistor, illuminated pushbuttons, fuses) is itemized in the component tables of the Hardware Design section and the point list in Appendix B.]

15.2 Appendix B: PLC I/O Point List

As-built I/O assignments for the DirectLOGIC 205 (D2-09B-1) base.

Rack slot layout

Slot	Module	Function
1	H2-DM1E	CPU with built-in Ethernet; serial COM port to HMI
2	D2-08ND3	8-ch discrete input, 24 VDC sourcing (X0–X7)
3	F2-08AD-1	8-ch analog current input; CH1 = VFD VO motor-current monitor
4	F2-04RTD	4-ch RTD input (all spare)
5	D2-08TR	8-ch relay output, sinking, common to PSU −V (Y0–Y7)
6	F2-08DA-2	8-ch analog voltage output; CH1 (+V1) = VFD VI speed reference
7–9	D2-Fill	Fill panels (reserved)

Discrete inputs (D2-08ND3, sourcing; common to PSU −V)

Addr	Field device / function
X0	Forward pushbutton (Autonics S2PR-P3G, NO)
X1	Reverse pushbutton (S2PR-P3Y, NO)
X2	Run 1 / Deadman 1 (S2PR-P3B, NO)
X3	Run 2 / Deadman 2 (S2PR-P3B, NO)
X4	E-stop monitor tap (E22B1 NC, in series with SD-N35 coil A1)
X5	VFD fault relay (FA–FC, NO active-high; HIGH on trip)
X6	Reserved – Reset pushbutton (S2PR-P3R, planned, unwired)
X7	Spare

Relay outputs (D2-08TR, sinking; common to PSU −V)

Addr	Load
Y0	Forward indicator LED (SA-LDG green)
Y1	Reverse indicator LED (SA-LDY yellow)
Y2	Run 1 / Deadman 1 LED (SA-LDB blue)
Y3	Run 2 / Deadman 2 LED (SA-LDB blue)
Y4	Error / fault indicator (Wamco 525, 28 VDC)
Y5	VFD RST (fault-reset pulse, ∼500 ms)
Y6	VFD FOR (forward run command)
Y7	VFD REV (reverse run command)

Analog and communications

Point	Connection
F2-08AD-1 CH1+	VFD VO terminal – motor-current monitor (0–10 V, PD054=1)
F2-08DA-2 +V1	VFD VI terminal – speed reference (0–10 V, PD002=1)
CPU serial COM	EA1-T4CL HMI (operator commands / status & data)
HMI power	Mean Well NDR-480-24 (+V / −V), 24 VDC

VFD terminals (Huanyang HY02D211B-T)

Terminal	Connection
R	AC hot via main disconnect + 30 A C-curve breaker
T	AC neutral via Phoenix Contact TMC 8 3C 15A breaker + SD-N35 contactor pole (as-built: neutral switched – see NEC note)
U / V / W	Motor T1/T2/T3 via colored banana sockets (blue/white/red)
P+ / PR	Brake resistor (ATO APCS-300R30-AD, 300 W, 30 Ω)
VI / VO	PLC F2-08DA-2 +V1 / PLC F2-08AD-1 CH1+
FOR / REV / RST	PLC D2-08TR Y6 / Y7 / Y5 (sinking, pulled to DCM)
FA / FC	PSU +V / PLC X5 (fault relay, PD052=02; FB unused)
DCM / ACM	PSU −V (shared digital / analog 0 V reference)

15.3 Appendix C: Wiring Diagrams

[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]

15.4 Appendix D: VFD Parameter Settings

The full programmed parameter set — code, function, value, and rationale — is given in Table 1 (Section 5.2, Motor and VFD). Motor values PD141–144 reflect the GE 5KE182BC205B test motor; the production motor is TBD and its nameplate will supersede these.

Power / braking wiring: R = AC hot (30 A breaker); T = AC neutral via SD-N35 contactor; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30 Ω brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

[NOTE: Firmware-specific finding: `PD052 = 12` (continue-running over-torque indication) did not fire the relay on our Huanyang revision; the working configuration is `PD052 = 02` with `PD123 = 3` (over-torque self-trips, which fires the fault relay and is read by the PLC on X5).]

15.5 Appendix E: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

ID	Test case
Top-down	Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY
BU-01	AC power input and fuse distribution
BU-02	24 VDC PSU output (Mean Well NDR-480-24)
BU-03	PLC rack initialization (DirectLOGIC 205)
BU-04	E-stop hardware safety circuit
BU-05	Two-hand start (deadman) safety logic
BU-06	Lid interlock (n/a in this build — access guarded mechanically by the ME team)
BU-07	VFD speed command and motor RPM (HY02D211B-T)
BU-08	Overcurrent / overload protection
BU-09	HMI operator interface (EA1-T4CL C-More Micro-Graphic)
BU-10	Status indicator lights (power / running / fault)

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

16 Authors and Acknowledgements

Authors

Bao Nguyen • **Fearghus Tyler** • **Yaqoub Rabiah** • **Fox Kang**
Department of Electrical & Computer Engineering, Portland State University.

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Disclaimer

This report documents an in-progress senior capstone design. Findings, parameters, and as-built details reflect the current state of the prototype and are subject to revision through final validation.

End of rough draft. Comments and questions welcome.